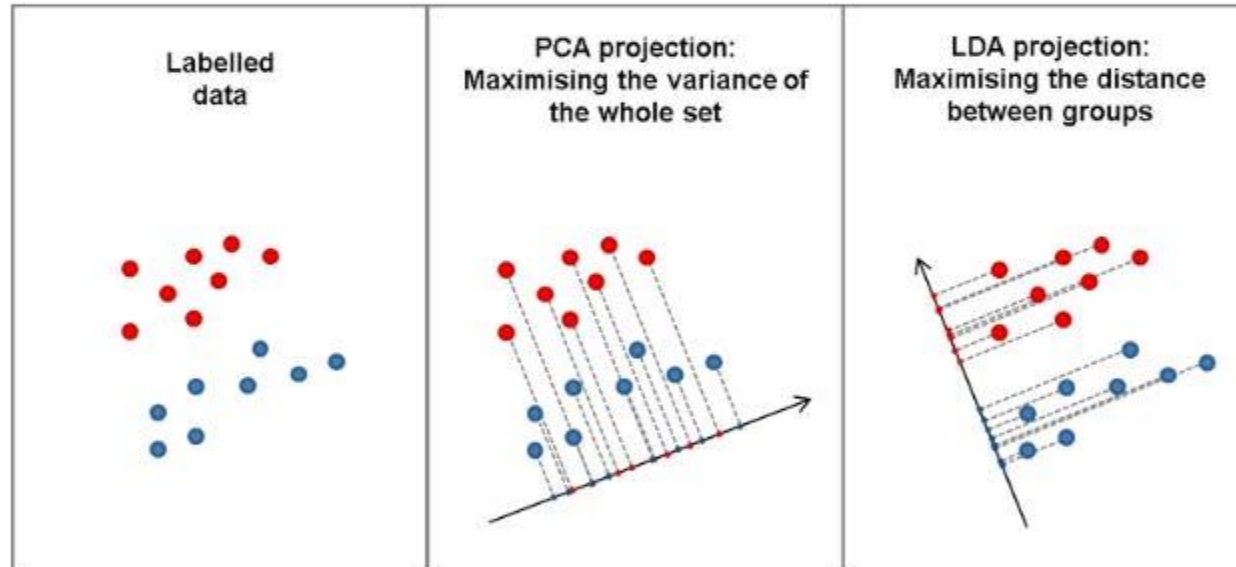
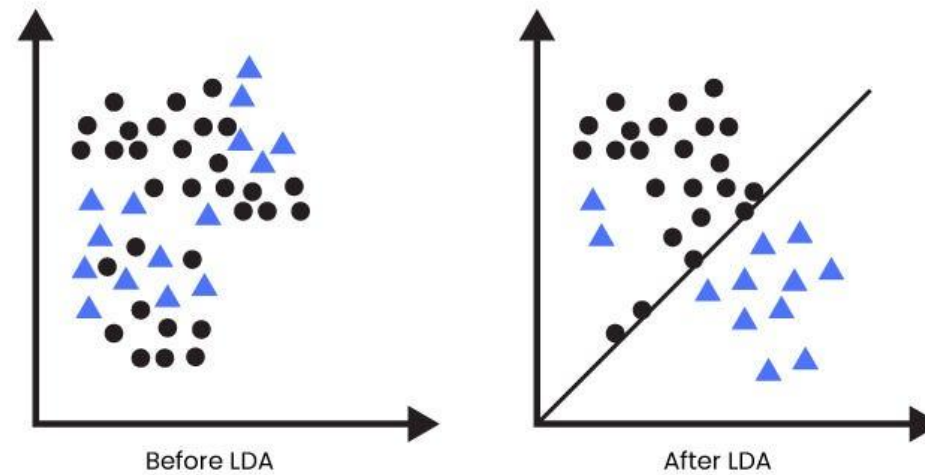


Linear Discriminant Analysis





LDA = maximize class separation.

Where is LDA used?

- Pattern recognition
- Face recognition
- Speech recognition
- Medical classification
- Image compression
- Finance (fraud detection)

Why LDA?

- Reduce dimensions while keeping class information
- Improve classification performance
- Remove noise
- Visualize high-dimensional data

LDA is a **supervised dimensionality reduction** and **classification** technique.

It finds a **new axis** (line) that **best separates different classes** by maximizing the distance between class means and minimizing the variance within each class.

“Find a linear transformation that best separates different classes.”

Unlike PCA (which focuses on variance),

LDA focuses on maximizing the separability between classes.

PCA	LDA
Both are dimensionality Reduction	Both are dimensionality Reduction
Both use EigenValue & Eigen Vector	Both use EigenValue & Eigen Vector
Compute maximum variance	Provide the separation boundary
Unsupervised – only features	LDA: Supervised – features + target



This gives maximum separation.

- Before LDA → Classes overlap in 2D
- After LDA → Data is projected onto a line that maximizes class separation.

LDA for Classification

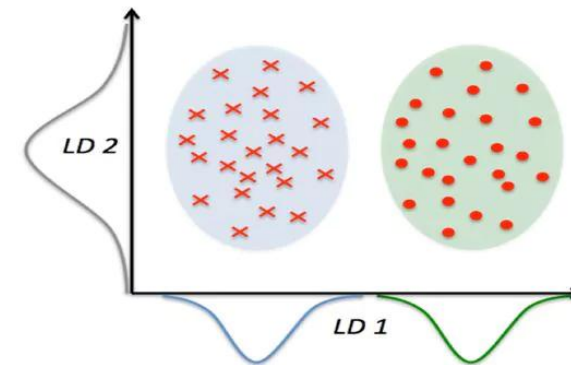
After projection, LDA assumes:

- Each class follows a **Gaussian distribution**
- All classes share a **common covariance matrix**

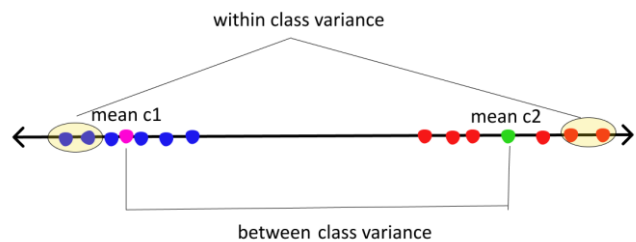
Classification is done using **Bayes theorem** on projected data.

After complete of LDA:

- Data for each class is **normally distributed**.
- Classes have **equal covariance matrices**.
- Classes are **linearly separable** in projected space.
- Features are **independent** (not strictly required).



Technique	Principal component analysis (PCA)	Independent component analysis (ICA)	Linear discriminant analysis (LDA)
Year	1990	1999	1997
Iterative	No	Yes	No
Class Information usage	No	No	Yes
Order of statistics	Second order	Higher order	Second order
Recognition rate (for 80 persons database)	70%	79%	89%
Speed	medium	very low	high
Scalability	low	low	high

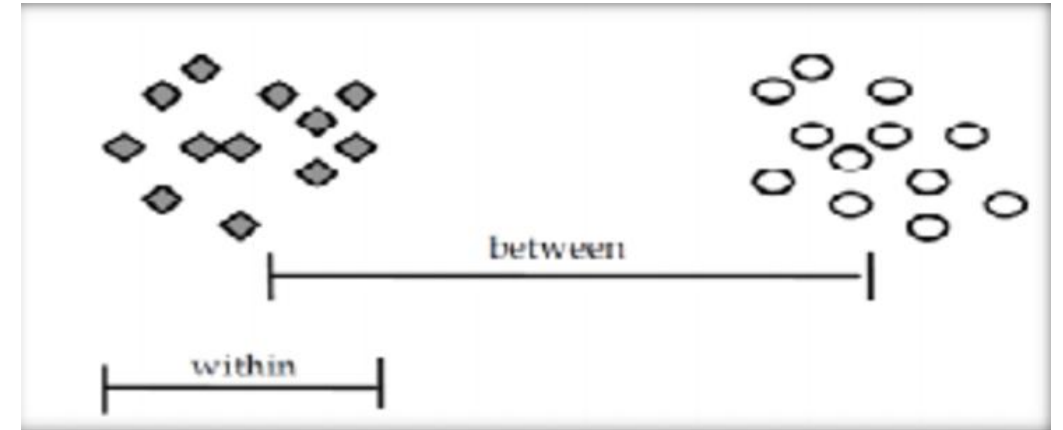
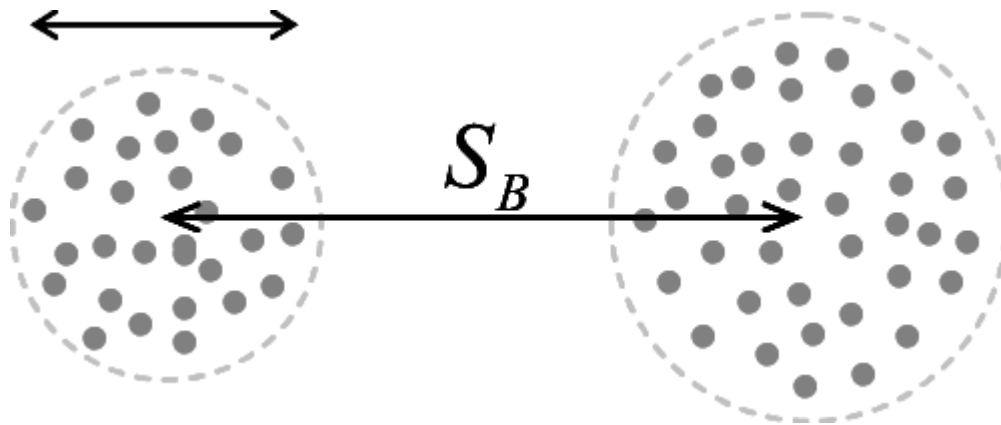


Advantages of LDA

- Simple and effective for classification.
- Works well with small datasets.
- Improves computation speed.
- Removes multicollinearity.
- Good for visualization.
- Better class separation than PCA.

Disadvantages of LDA

- Assumes Gaussian distribution.
- Fails when classes have:
 - Different covariance matrices
 - Non-linear boundaries
- Sensitive to outliers.
- Poor performance with heavily overlapping classes.



Mathematical Steps of LDA

- We have data X with n samples and d features
- There are k classes
- Each class is $C_1, C_2, \dots, C_k$

STEP 1 — Compute Class Mean Vectors

For each class i :

$$\mu_i = \frac{1}{N_i} \sum_{x \in C_i} x$$

Also compute overall mean:

$$\mu = \frac{1}{N} \sum_{i=1}^k \sum_{x \in C_i} x$$

STEP 2 — Compute Within-Class Scatter Matrix S_W

$$S_W = \sum_{i=1}^k \sum_{x \in C_i} (x - \mu_i)(x - \mu_i)^T$$

STEP 3 — Compute Between-Class Scatter Matrix S_B

$$S_B = \sum_{i=1}^k N_i (\mu_i - \mu)(\mu_i - \mu)^T$$

STEP 4 — Solve the Generalized Eigenvalue Problem

$$S_W^{-1} S_B w = \lambda w$$

- w = eigenvectors $\rightarrow$ LDA directions
- λ = eigenvalues $\rightarrow$ importance of each LDA component

Sort eigenvectors by decreasing eigenvalues.

STEP 5 — Select Top Components

Max LDA components = $k - 1$

$$W = [w_1, w_2, \dots, w_{k-1}]$$

STEP 6 — Project the Data

$$X_{\text{LDA}} = XW$$

This gives LDA1, LDA2, ...

Mathematical intuition with Example

Dataset

Let the classes be:

Class 1 (C_1)

Sample	x_1	x_2
1	4	2
2	2	4
3	2	3

Class 2 (C_2)

Sample	x_1	x_2
1	7	8
2	8	8
3	9	10

STEP 1: Compute Class Means

Class 1 mean

$$\mu_1 = \left(\frac{4 + 2 + 2}{3}, \frac{2 + 4 + 3}{3} \right)$$
$$\mu_1 = (2.67, 3)$$

Class 2 mean

$$\mu_2 = \left(\frac{7 + 8 + 9}{3}, \frac{8 + 8 + 10}{3} \right)$$
$$\mu_2 = (8, 8.67)$$

✓ STEP 2: Compute Within-Class Scatter Matrix S_W

Formula:

$$S_W = \sum (x - \mu_i)(x - \mu_i)^T$$

We compute it for each sample.

● Class 1 Samples

Sample 1: (4,2)

$$x - \mu_1 = (4 - 2.67, 2 - 3) = (1.33, -1)$$

Scatter:

$$\begin{bmatrix} 1.33^2 & 1.33(-1) \\ 1.33(-1) & (-1)^2 \end{bmatrix} = \begin{bmatrix} 1.77 & -1.33 \\ -1.33 & 1 \end{bmatrix}$$

Sample 2: (2,4)

$$(2 - 2.67, 4 - 3) = (-0.67, 1)$$

Scatter:

$$\begin{bmatrix} 0.45 & -0.67 \\ -0.67 & 1 \end{bmatrix}$$

Sample 3: (2,3)

$$(2 - 2.67, 3 - 3) = (-0.67, 0)$$

Scatter:

$$\begin{bmatrix} 0.45 & 0 \\ 0 & 0 \end{bmatrix}$$

In Details Matrix Operation

$$(x - \mu)(x - \mu)^T$$

🎯 Step 1: Write the sample and the class mean

Suppose a sample:

$$x = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Class mean:

$$\mu = \begin{bmatrix} 2.67 \\ 3 \end{bmatrix}$$

Compute the difference:

$$x - \mu = \begin{bmatrix} 4 - 2.67 \\ 2 - 3 \end{bmatrix} = \begin{bmatrix} 1.33 \\ -1 \end{bmatrix}$$

This vector is of shape (2×1).

🎯 Step 2: Compute the outer product

Scatter matrix definition:

$$S = (x - \mu)(x - \mu)^T$$

This is matrix multiplication of:

- A (2×1) column vector
- With its (1×2) row transpose

So:

$$\begin{bmatrix} 1.33 \\ -1 \end{bmatrix} \begin{bmatrix} 1.33 & -1 \end{bmatrix}$$

Final scatter matrix of this sample

$$S = \begin{bmatrix} 1.77 & -1.33 \\ -1.33 & 1 \end{bmatrix}$$

● Class 2 Samples

Sample 1: (7,8)

$$(7 - 8, 8 - 8.67) = (-1, -0.67)$$

Scatter:

$$\begin{bmatrix} 1 & 0.67 \\ 0.67 & 0.45 \end{bmatrix}$$

Sample 2: (8,8)

$$(8 - 8, 8 - 8.67) = (0, -0.67)$$

Scatter:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0.45 \end{bmatrix}$$

Sample 3: (9,10)

$$(9 - 8, 10 - 8.67) = (1, 1.33)$$

Scatter:

$$\begin{bmatrix} 1 & 1.33 \\ 1.33 & 1.77 \end{bmatrix}$$

◆ Total Class 2 scatter

Add all:

$$S_{W2} = \begin{bmatrix} 1 + 0 + 1 & 0.67 + 0 + 1.33 \\ 0.67 + 0 + 1.33 & 0.45 + 0.45 + 1.77 \end{bmatrix}$$

$$S_{W2} = \begin{bmatrix} 2 & 2 \\ 2 & 2.67 \end{bmatrix}$$

■ Total Within-class Scatter

$$S_W = S_{W1} + S_{W2}$$

$$S_W = \begin{bmatrix} 2.67 + 2 & -2 + 2 \\ -2 + 2 & 2 + 2.67 \end{bmatrix}$$

$$S_W = \begin{bmatrix} 4.67 & 0 \\ 0 & 4.67 \end{bmatrix}$$

This is diagonally nice 🍌.

✓ STEP 3: Compute Between-Class Scatter S_B

Formula:

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

Compute difference:

$$d = \mu_1 - \mu_2 = (2.67 - 8, 3 - 8.67) = (-5.33, -5.67)$$

Scatter:

$$S_B = \begin{bmatrix} 5.33^2 & 5.33 \cdot 5.67 \\ 5.33 \cdot 5.67 & 5.67^2 \end{bmatrix}$$
$$S_B = \begin{bmatrix} 28.4 & 30.2 \\ 30.2 & 32.1 \end{bmatrix}$$

✓ STEP 4 — Solve the Generalized Eigenvalue Problem

LDA solves:

$$S_W^{-1} S_B w = \lambda w$$

■ Step 4.1 — Compute S_W^{-1}

Since S_W is diagonal:

$$S_W^{-1} = \begin{bmatrix} 1/4.67 & 0 \\ 0 & 1/4.67 \end{bmatrix}$$

Numeric value:

$$1/4.67 \approx 0.214$$

So:

$$S_W^{-1} = \begin{bmatrix} 0.214 & 0 \\ 0 & 0.214 \end{bmatrix}$$

■ Step 4.2 — Multiply $S_W^{-1} S_B$

$$S_W^{-1} S_B = \begin{bmatrix} 0.214 & 0 \\ 0 & 0.214 \end{bmatrix} \begin{bmatrix} 28.40 & 30.20 \\ 30.20 & 32.10 \end{bmatrix}$$

Multiply row-by-column:

First row:

$$0.214 \cdot 28.40 = 6.08$$

$$0.214 \cdot 30.20 = 6.46$$

Second row:

$$0.214 \cdot 30.20 = 6.46$$

$$0.214 \cdot 32.10 = 6.87$$

So:

$$A = S_W^{-1} S_B = \begin{bmatrix} 6.08 & 6.46 \\ 6.46 & 6.87 \end{bmatrix}$$

Step 4.3 — Solve Eigenvalue Problem

We solve:

$$Aw = \lambda w$$

We find eigenvalues of:

$$\begin{vmatrix} 6.08 - \lambda & 6.46 \\ 6.46 & 6.87 - \lambda \end{vmatrix} = 0$$

Compute determinant:

$$(6.08 - \lambda)(6.87 - \lambda) - (6.46)^2 = 0$$

Expand:

$$(6.08)(6.87) - 6.08\lambda - 6.87\lambda + \lambda^2 - 6.46^2 = 0$$

Numerical values:

- $6.08 \cdot 6.87 = 41.77$
- $6.46^2 = 41.73$

So:

$$\lambda^2 - (6.08 + 6.87)\lambda + 41.77 - 41.73 = 0$$

$$\lambda^2 - 12.95\lambda + 0.04 = 0$$

Solve quadratic:

$$\lambda = \frac{12.95 \pm \sqrt{12.95^2 - 0.16}}{2}$$

$$12.95^2 = 167.70$$

$$\sqrt{167.70 - 0.16} = \sqrt{167.54} = 12.94$$

Thus:

Eigenvalue 1:

$$\lambda_1 = \frac{12.95 + 12.94}{2} = 12.945$$

Eigenvalue 2:

$$\lambda_2 = \frac{12.95 - 12.94}{2} = 0.005$$

Step 4.4 — Find eigenvector for LDA direction

We take eigenvector of **largest eigenvalue** (12.945).

Solve:

$$(A - 12.945I)w = 0$$

Plug in:

$$\begin{bmatrix} 6.08 - 12.945 & 6.46 \\ 6.46 & 6.87 - 12.945 \end{bmatrix} = \begin{bmatrix} -6.865 & 6.46 \\ 6.46 & -6.075 \end{bmatrix}$$

Solve system:

$$-6.865w_1 + 6.46w_2 = 0$$

$$6.46w_1 - 6.075w_2 = 0$$

Use first equation:

$$6.46w_2 = 6.865w_1$$

$$\frac{w_2}{w_1} = \frac{6.865}{6.46} = 1.062$$

Take:

- $w_1 = 1$
- $w_2 = 1.062$

So eigenvector:

$$w = \begin{bmatrix} 1 \\ 1.062 \end{bmatrix}$$

Normalize:

Magnitude:

$$\sqrt{1^2 + (1.062)^2} = \sqrt{1 + 1.128} = \sqrt{2.128} = 1.46$$

So:

$$w = \frac{1}{1.46} \begin{bmatrix} 1 \\ 1.062 \end{bmatrix} = \begin{bmatrix} 0.685 \\ 0.727 \end{bmatrix}$$

LDA Direction Vector

$$w = \begin{bmatrix} 0.685 \\ 0.727 \end{bmatrix}$$

This is the **optimal projection direction** that maximizes class separation.

✅ STEP 5 — Project the data

Projection formula:

$$y = w^T x$$

Example for sample $x = (4, 2)$:

$$y = 0.685(4) + 0.727(2)$$

$$y = 2.74 + 1.454 = 4.194$$

This is its new coordinate on **LDA1 axis**.

Do this for all samples → you get the new 1D representation.